

Experiment 8: Nmap – Service Version & OS Detection

Aim

To perform service version detection, aggressive scanning and operating system detection using Nmap:

- -sV - Detect versions of services running on open ports
- -A - Aggressive scan (OS detection + version + traceroute + scripts)
- -O - Operating System detection

Procedure

1. Open Terminal in Kali Linux.
2. Detect service versions on a target (use scanme.nmap.org or your lab target):
`nmap -sV scanme.nmap.org`
3. Perform an aggressive scan:
`nmap -A scanme.nmap.org`
4. Detect the operating system of the target:
`nmap -O scanme.nmap.org`
5. Note: Always scan only systems you own or have permission to test.

Sample Output / Results

i) Service Version Detection (-sV):

```
$ nmap -sV -p 80 scanme.nmap.org
PORT      STATE SERVICE VERSION
80/tcp    open  http    Apache httpd 2.4.7 ((Ubuntu))
Service detection performed...
```

ii) Aggressive Scan (-A):

```
$ nmap -A -p 80 scanme.nmap.org
PORT      STATE SERVICE VERSION
80/tcp    open  http    Apache httpd 2.4.7 ((Ubuntu))
|_http-title: Go ahead and ScanMe!
Aggressive OS guesses: Linux 4.15 - 5.8 (95%)
Network Distance: 16 hops
```

iii) OS Detection (-O):

```
$ nmap -O scanme.nmap.org
Aggressive OS guesses:
Linux 4.15 - 5.8 (95%), Linux 5.0 - 5.4 (95%)
No exact OS matches for host (test conditions non-ideal).
```

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ nmap -sV scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-03 12:24 -0400  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.0059s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 987 filtered tcp ports (no-response)  
PORT      STATE SERVICE      VERSION  
22/tcp    open  tcpwrapped  
53/tcp    closed domain  
80/tcp    open  http          Apache httpd 2.4.7 ((Ubuntu))  
139/tcp   closed netbios-ssn  
143/tcp   closed imap  
443/tcp   closed https  
500/tcp   closed isakmp  
3389/tcp  closed ms-wbt-server  
5002/tcp  closed rfe  
5060/tcp  open  sip?  
5061/tcp  open  sip-tls?  
8080/tcp  closed http-proxy  
8081/tcp  closed blackice-icecap  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 107.37 seconds  
  
(kali@kali)-[~]  
$ nmap -A scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-03 12:27 -0400  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.00089s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
Not shown: 988 filtered tcp ports (no-response)  
PORT      STATE SERVICE      VERSION  
22/tcp    open  tcpwrapped  
|_ssh-hostkey: ERROR: Script execution failed (use -d to debug)  
53/tcp    closed domain  
80/tcp    open  http          Apache httpd 2.4.7 ((Ubuntu))  
|_http-server-header: Apache/2.4.7 (Ubuntu)  
139/tcp   closed netbios-ssn  
143/tcp   closed imap  
443/tcp   closed https  
993/tcp   closed imaps  
3389/tcp  closed ms-wbt-server  
5060/tcp  open  sip?  
5061/tcp  open  sip-tls?  
8080/tcp  closed http-proxy  
8085/tcp  closed unknown  
Device type: VoIP adapter|bridge|general purpose  
Running (JUST GUESSING): AT&T embedded (90%), Oracle Virtualbox (88%), Slirp (88%), QEMU (86%)  
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
```

```
Device type: voip adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (90%), Oracle Virtualbox (88%), Slirp (88%), QEMU (86%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (90%), Oracle Virtualbox Slirp NAT bridge (88%), QEMU user mode ne
twork gateway (86%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
```

```
TRACEROUTE (using port 80/tcp)
HOP RTT      ADDRESS
1   0.25 ms scanme.nmap.org (45.33.32.156)
```

```
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 134.13 seconds
```

```
(kali㉿kali)-[~]
└─$ sudo nmap -O scanme.nmap.org
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-03 12:29 -0400
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.012s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 989 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    closed domain
80/tcp    open  http
139/tcp    closed netbios-ssn
143/tcp    closed imap
161/tcp    closed snmp
443/tcp    closed https
3389/tcp   closed ms-wbt-server
5060/tcp   open  sip
5061/tcp   open  sip-tls
8080/tcp   closed http-proxy
OS fingerprint not ideal because: Didn't receive UDP response. Please try again with -sSU
No OS matches for host

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 37.87 seconds
```